

ANSWER KEY

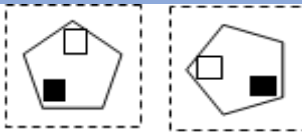
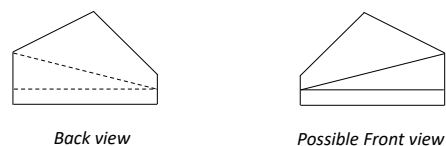
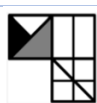
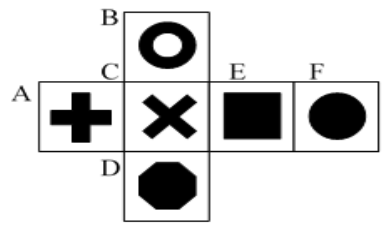
I. LOGICAL REASONING

VERBAL REASONING

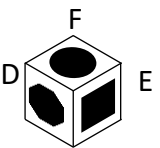
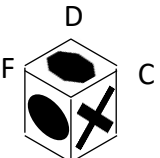
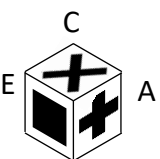
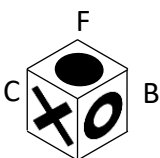
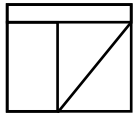
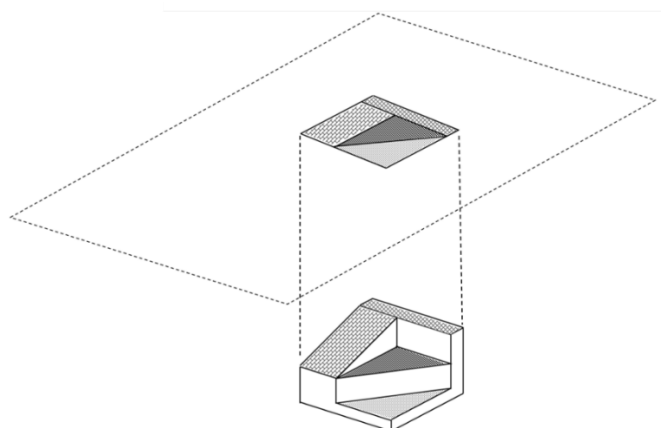
ITEM NO.	ANSWER	RATIONALIZATION
1	C	
2	A	
3	A	
4	A	
5	A	
6	C	
7	B	
8	C	
9	D	
10	D	

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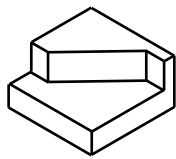
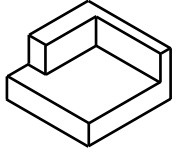
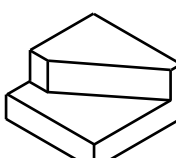
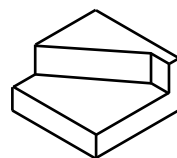
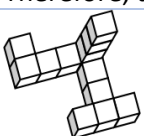
NON-VERBAL REASONING

ITEM NO.	ANSWER	RATIONALIZATION
1	C	 Rotate Left 90°
2	D	When the other options are spatially rotated, they are a mirror image of the target image.
3	D	Each diagram can be seen as a series of triangles around a horizontal axis. At each step, the right-hand triangle relocates to the beginning of the sequence causing a shift to the right.
4	D	All frames besides frame D are symmetrical, which means that they can be crossed somewhere by a straight line that will divide them into two similar parts (mirrored). There is no such option in frame D.
5	A	The back view can be converted into a front view by flipping and making the hidden lines visible:  <i>Back view</i> <i>Possible Front view</i>
6	B	
7	C	
8	C	
9	C	
10	B	
11	D	 It is the original block turned once clockwise. The circle faces forward. Next, it is turned on its side so that the diamond moves from the top to the right side.
12	D	 The logic: Pairs of columns are reflections of one another.
13	A	The arrangement of the cube in a flat surface when the two pieces of cube puzzle is combined is:  For the view cube, possible combinations of the views are: ABC; ACD; BCE; CDE; BEF; DEF; ABF and ADF

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		For the choices: A.  For sides D and F, the only possible third side is E or A. Therefore, it is a correct view cube B.  For sides D and F, the only possible third side is E or A. Therefore, it is a wrong view cube C.  For sides E and C, the only possible third side is B or D. Therefore, it is a wrong view cube D.  For sides B and F, the only possible third side is A or E. Therefore, it is a wrong view cube Therefore, the answer is A.
14	B	The shaded parts are the surfaces that can be projected and be viewed on the top plane. And the projected plane is the same as figure at letter B.  Therefore, the answer is B. 

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15	A	A.  Front view: A rectangle with a vertical line on the left side. Right side view: A rectangle with a vertical line on the right side. B.  Front view: A rectangle with a vertical line on the left side. Right side view: A rectangle with a vertical line on the right side. C.  Front view: A rectangle with a vertical line on the left side. Right side view: A rectangle with a vertical line on the right side. D.  Front view: A rectangle with a vertical line on the left side. Right side view: A rectangle with a vertical line on the right side. Therefore, the answer is A.
16	B	
17	D	
18	D	
19	B	
20	D	

ANSWER KEY

II. POWER TEST

ENGLISH

ITEM NO.	ANSWER	RATIONALIZATION
1	B	The answer is “Roman’s Math scores have improved”. The verb phrase should be in the present perfect tense with the Auxiliary Verb in the plural form (have) to agree with the subject “scores.” Roman’s scores improved from 85% to 87%. The time reference is last quarter (some point in the past) and this quarter (some point in the present). In this context, the appropriate verb tense is Present Perfect which indicates an action that occurred over a prior time period and that is completed at the moment of speaking. This captures the progress of Roman’s scores over time.
2	A	The answer is “A report on renewable energy was submitted by a group of students”. The item requires a rephrasing of the stem which is in active voice to its passive form. To transform into passive, the object of the verb (report on renewable energy) must be the subject of the sentence. This condition is fulfilled by all the options; however, the verb form must also be accurate. The required verb is in the simple past passive form (was submitted) because the tense in the stem is simple past and the object turned subject (A report) is singular.
3	B	The answer is “would not have happened”. This item requires the proper verb form in a Type 3 conditional sentence (untrue in the past). If the if-clause is in the past perfect (had been more careful), the result clause should use would have + past participle (would not have happened).
4	C	The answer is “Girls’ bond over sharing secrets and boys’ preference for doing activities together show gender differences when children socialize”. To achieve an adequate paraphrase, these are some steps that can be considered: (1) locate the major idea units, (2) change the order of major idea units but maintain their logical connection, and (3) substitute synonyms for some words in the original. All options fulfill step (1) because both major idea units are mentioned: that there is gender difference in childhood socialization and that girls and boys have different preferences in establishing social connections. However, it is only Option C which fulfills all three steps to achieve an adequate paraphrase maintaining the logical connection indicated in the original sentence.
5	B	The answer is “Technology-mediated communication has changed how people use various media to build relationships”. Options A, C, and D are the same in expressing “technology-mediated communication through various media” as one idea unit which changed “how people build relationships.” Option B is different in this aspect because

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		“use of various media” is clustered with “people” “how people use various media.”
6	C	The answer is “Anyone is capable of leadership which is defined as influence”. Option A is partly true about the passage. Option D is logical based on one’s background knowledge. Option B is incorrect as the passage mentions that leadership is not about power and charisma. Option C provides a good gist of the passage which states how anyone can be a leader given that he/she can influence others.
7	A	The answer is Use “Moreover” to begin sentence 3 to improve coherence. The paragraph does not have any problem in content and unity. The inclusion of a cohesive device improves the logical connection of ideas in the paragraph. The spider’s silk elasticity is an additional detail, thus the use of “Moreover.”
8	C	The answer is “To conclude the paragraph”. Sentence 8: The wise use of telescopes made us better understand our universe is a concluding sentence.
9	B	The answer is “Scientists used telescopes to learn about planets”. Options C and D are specific details that do not capture the major idea about telescopes. Option A is an illogical statement since the scientists and not the telescopes who studied gravity and speed of light. Option B captures the main idea about telescopes being a useful tool in studying planets and space.
10	A	The answer is “Sentences 3 and 5”. The item requires supporting details. Sentences 3 and 5 provide these because they are about specific important discoveries: moons orbit around Jupiter and the sun is the center of the universe. Sentences in Option B are main idea sentences. Options C and D combine less important ideas that do not directly support the main idea sentence: The use of telescopes led to important discoveries.
11	D	The answer is “comparison-contrast”. The topic sentence indicates that the paragraph compares renewable energy and fossil fuels. The use of signal words including “more,” “in contrast,” and “while” are also clues that show the pattern of paragraph development. Examinees may choose option A since some of the supporting details express effects of renewable energy and fossil fuels to the environment. However, this is just one point of comparison used in the paragraph.
12	C	The answer is “The use of renewable energy benefits people and the environment”. Options A and B are specific points in the paragraph, but Option C captures the main idea that should be emphasized to highlight the benefits of renewable energy.

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13	D	The answer is “Show a comparison of renewable energy and fossil fuels”. The pattern of paragraph development is comparison and contrast, so it will make sense to also present the information in the infographic the same way. Options A will only feature certain specific details from the paragraph, while Options B and C are details not present in the paragraph.
14	A	“Varsity” is singular; the verb should be “wins”
15	A	“Ham and cheese” is taken as a pair and considered singular; verb should be “is”
16	C	“I” does not follow regular subject-verb agreement rules; it takes a plural verb form (like the singular form of “you”). The correct verb here should be “sing”
17	D	It should be recognized that this is an idiomatic expression item, i.e., a blessing in disguise = an apparent misfortune that eventually has good results.
18	D	This is a rephrasing item. Letter D retained the emphasis of confidence of the original statement in passing the test, and the cause-effect relationship is clear.
19	D	This is a logical connectors item. Yet is a coordinating conjunction which uses the “nevertheless” sense.
20	D	This is a vocabulary in context item. C and D get points, but “margin” is similar to “periphery” in social structure contexts (which is what the article is about), while flank is not.
21	D	This is a “predict a possible objection to a benefit” item. C gets less points than D because although rebellion seems bigger than scamming, rebellion could be justified and could hardly be staged on mobile phones (alone) while scamming is not, and it’s disadvantage is always personal.
22	C	“3” is the most relevant purpose among the options as it really is the meat of the article (mobilizing people), Letter C is the best answer because “2” is the second most relevant purpose because it is about two types of mobilizing people.
23	D	This is a textual evidence item, and the best answer can be found in the article itself. The best answer is D.
24	C	
25	B	
26	C	
27	D	
28	D	
29	C	
30	D	
31	D	
32	B	
33	A	
34	A	
35	A	
36	D	
37	D	
38	A	“Eulogy” in the title already provides a clue in the poem’s main idea. Also, the concept of death applies not only to people but to roaches as well. C and

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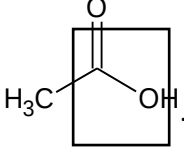
		D get a point each because they support the main idea, while B does not get a point because it is too general, and it does not apply to the roaches.
39	B	This is a vocabulary in context item. A and B are similar in meaning, but “accolade” is not used to refer to the dead, while “tribute” is. C and D get zero points because they are entirely different concepts from eulogy.
40	B	This item determines the stand or tone of the writer in the article. A, B, and D get zero points because they are not apparent in the poem at all. B is the correct answer because the writer presents the idea of death matter-of-factly or dis-passionately.
41	D	This is a textual evidence item, and the best answer can be found in the poem itself. The best answer is D.
42	D	This is an “applying concepts to practical situations” item. A and C are acceptable answers, and they are both ideally organized. D is the best answer because it includes mostly the important parts of an email in an ideal order. B gets zero points because CC/BCC are not that necessary—one could always send the same email directly to those recipients.
43	C	This is a vocabulary in context item. C is the best answer because it is like the idea of eviction—an indefinite departure initiated by an external force. A and B are similar because they could be both voluntary actions. D is initiated by an external factor but only for a specific period.

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SCIENCE

ITEM NO.	ANSWER	RATIONALIZATION
1	A	Magnification is the product of the magnification of the eyepiece and the objective used. To compute for the magnification of the objective, you should divide the overall magnification with that of the eyepiece (100 / 20).
2	C	To photosynthesize, plants need light, water, and carbon dioxide. They do not need oxygen because oxygen is a by-product of photosynthesis.
3	B	Animals that live in tundra should have rich fat reserves that they need to nourish their bodies during their hibernation period. They also need a thick fur coat to insulate them from the extreme cold temperature in this biome.
4	B	
5	D	
6	A	
7	C	The fine adjustment knob is adjusted to sharpen the focus. The coarse adjustment knob is to adjust the distance between the objectives and the slide to look for the specimen. The eyepiece is where you view the image. The mirror serves as a light source.
8	A	Pag-asa is a cactus. Cacti can live in deserts because they have stores of water in their succulent stems and they have reduced leaf surface area to lessen water loss through transpiration.
9	C	According to the Kinetic Molecular Theory, the molecules move faster when the temperature is raised, thereby increasing their KEs. Hence, option C is false.
10	A	Refer to rules in chemical nomenclature. In general, the element should be properly represented by the appropriate chemical symbol in the Periodic Table. The cation's charge becomes the anion's subscript, while the anion's charge becomes the cation's subscript. The subscripts can be taken as a ratio and can be deduced to simpler forms. Only A follows these general rules.
11	A	All of the given choices, except C_3H_8 are ionic compounds since they are formed by the complete transfer of electrons from a metal to a non-metal. The covalent compound C_3H_8 is formed from sharing of electrons between two non-metals.
12	B	Copper is paramagnetic and can be recovered using a magnet. Water can be easily removed by filtration or decantation. Distillation is a process used to separate two miscible liquids, and not two solids. Hence, method B will separate the mixture most effectively.
13	D	The separation of the pigments from the solution can best be achieved via chromatography, since the basis of separation is polarity. Centrifugation would definitely not separate the pigments from the solution since they are both in liquid form. Distillation and evaporation may do, but not as effective as chromatography.
14	D	Sodium chloride, NaCl, or table salt, is a solid compound at room temperature. It is an ionic compound that conducts electricity when dissolved in water.
15	A	The best answer is letter A which is that of line segments A and D. It is because in a displacement versus time graph. A horizontal line segment represents no displacement. This means that the object, in this case the grab

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		taxi is not moving. Graphs A and D are horizontal line segments representing no movement. Line segment B a graph with positive slope. This graph shows a constant speed moving away from the original position. Line segment B on the other hand represents a constant speed with the object returning to the original position. Ray E represents a changing speed, meaning that the graph is accelerating.
16	C	
17	A	
18	A	
19	B	
20	C	
21	B	Definition of Comet.
22	D	During a total solar eclipse, the moon is between the sun and the earth. If the earth is between the sun and the moon, it is a lunar eclipse. A and B are impossible since the moon rotates around the earth.
23	B	Sun moves east to west. If you face north, the sun will move from right to left.
24	A	Clastic sedimentary rocks are formed when pieces of pre-existing rocks are loosened by weathering and transported to other places where some sediments are trapped and hardened.
25	A	Decreasing the slope and water running smoothly slow down the mass movement of sediments. These are the factors in the formation of delta.
26	D	
27	D	
28	B	
29	A	
30	B	
31	A	Definition of an Asteroid.
32	A	Magnification is the product of the magnification of the eyepiece and the objective used. To compute for the magnification of the objective, you should divide the overall magnification with that of the eyepiece (100/10).
33	C	When fish need some salt to live, this means that they have evolved in a marine environment. Estuaries are brackish areas where saltwater meets the freshwater coming from mouth of rivers.
34	B	
35	C	Carboxylic acids have the functional group $\text{R}-\text{C}(=\text{O})\text{OH}$. This is present in acetic acid as shown below 
36	C	It has the balanced chemical equation with correct formula for the reactants and products.

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37	D	It shows the correct sequence of steps in determining the empirical formula of the compound.
38	C	Follow Aufbau's Principle in writing the electronic configuration. If the element has a nuclear charge of +16, the element also has 16 electrons. Also, bear in mind that s can hold a maximum of 2 electrons, p can hold 6, d can hold 10 and f can hold 14. Hence, the electron configuration should be $1s^2 2s^2 2p^6 3s^2 3p^4$, which is letter C.
39	C	Solve for the proton number (P), neutron number (N) and electron number (E), independently. For an atom of the notation A_ZX , recall that the atomic number, Z, is equivalent to the proton number and electron number. The mass number A is equivalent to the sum of the proton number and neutron number. Hence, for ${}^{14}_7\text{N}$ $Z = P = E = 7$ The neutron number, N, is $N = A - P = 14 - 7 = 7$ Nitrogen has the same number of protons, neutrons and electrons.
40	D	Solve for the proton number (P), neutron number (N) and electron number (E), independently. For an atom of the notation A_ZX , recall that the atomic number, Z, is equivalent to the proton number and electron number. The mass number A is equivalent to the sum of the proton number and neutron number. Hence, for ${}^{16}_8\text{O}$ $Z = P = E = 8$ The neutron number, N, is $N = A - P = 16 - 8 = 8$ Oxygen has the same number of protons, neutrons and electrons.
41	B	
42	B	The best answer is letter B. For maximum range to be attained, an angle for 45 degrees should be considered when firing a projectile with respect to the ground. It is derived from the equation for the range of a projectile given an initial velocity: $R = \frac{v_o^2(\sin 2\theta)}{g}$ where: R – range – horizontal displacement v_o - initial velocity θ – angle with respect to the horizontal g – gravitational acceleration constant (9.81 m/s^2) If 45 will be substituted in the equation, then the $\sin 2\theta$ part of the equation will yield a maximum value of 1. This means that this becomes the maximum value of the angle to reach the farthest horizontal displacement from the original position.
43	D	The best Answer is letter D. We consider each of the options and tell which are not reasonable and which involves a change in speed. I. Fifteen minutes of break time spent eating in a table at the canteen.

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		This activity requires a change of speed as Ben may be moving and then stops or is initially not moving then walks. A time element of fifteen minutes for break is also reasonable. This means that this is not an option that we are looking for. II. Twenty minutes to walk to the nearby lecture room in Science. This activity is not reasonable. A nearby canteen will not take more than 20 minutes to go to. Spending more than 10 minutes will not be reasonable. This activity also involves possible variations in speed of Ben. This means that this fits the description in what is asked in the problem. III. Two hours to attend the Science lecture-laboratory. First, this activity does not involve a change in speed. Attending a lecture usually requires students to be seated and are therefore not moving. Spending two hours in a lecture is also acceptable. IV. Twenty-minute walk to a nearby canteen. Walking involves possible changes in speed. Spending twenty minutes to walk to a “nearby” canteen is also not reasonable. Canteens are usually near and does not require more than 10 minutes to go to. V. An hour to attend a lecture in Math. Attending an hour of lecture usually does not require a change in speed. It also usual that lecture sessions to be around an hour. This means that only options II and IV fit what is required by the item.
44	C	The best answer is letter C. The final displacement simply is determined by determining the distance of the object from its initial position to its final position. As can be seen in the graph, you only need to focus on graph 8 which shows that the object is 8 meters away from the initial position. This means that its displacement is 8 meters.
45	C	The best Answer is letter C. We consider each of the options and tell which are not reasonable and which involves a change in speed. I. Two hours bus nonstop trip from home to the church 70 m away. Spending more than 1 hour to cover 70 meters is <u>not reasonable</u>. A bus can travel at an acceptable rate of 80 km per hour and as slow as more or less than 20 km per hour. Even if the bus travels at 1 km per hour, the bus will surely cover the distance of 70 meters in less than an hour. II. An hour spent to attendance the mass. It is reasonable for a person to spend an hour stationary in the church. A usual church service covers an hour. III. One-minute walk to the market 1 km away from the church;

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		It is impossible for a person to cover a kilometer by walking. Even if a person covers 2 meters per step, the person can only cover 120 meters in a minute. This means that the speed of the person needs some revisions. IV. Thirty minutes spent to buy goods from Aling Susan's grocery store in the market. Spending thirty minutes to buy goods is reasonable. This means that only options I and III are NOT reasonable and the options asked in this item. The option that refers to these is option C.
46	B	The best answer is B. A horizontal graph in a velocity versus time graph represents constant velocity. A slanted graph represents an accelerating object. If the slant has a positive slope, the acceleration is positive and if the slant has a negative slope, the acceleration is negative.
47	A	Sequence from oldest to youngest (bottom to top) 1. Siltstone P 2. Shale M 3. Limestone C 4. Sandstone L 5. Limestone D 6. Sandstone Q 7. Siltstone O
48	C	Absolute dating uses the isotopes present in a fossil to know the age of the fossil. Carbon dating is used for fossils that are organic and are up to 10,000 years old. Potassium-Argon is used for older fossils.
49	C	
50	B	
51	B	All living organisms make an impact on soil development. Humus is formed in the upper layer of the soil, which is composed of dirt, litter, water and air. Soil does not form overnight. A variety of micro-organisms in the soil help decompose living materials, making the soil rich in nutrients. Increasing the amount of living organism in soil increases the biological activity. A, C and D. Slope and air pressure have no impact on soil development, decreasing the amount of rainfall inhibits the development of the soil.
52	D	There are two types of energy source – renewable and non-renewable. Renewable energy source can be replenished right away, while non-renewable can be replenished for a longer period. Knowing these definitions, one must know how to conserve energy. Each one should know their role/s in utilizing energy.
53	C	
54	C	Earth's layers from outside-in: crust, mantle, outer core, inner core. With boundaries/discontinuities: crust, Moho, mantle, Gutenberg, outer core, Lehmann, inner core. Boiled egg: shell, albumen, yolk (core).
55	D	Silt can be described as loosely packed suspended sediments. It is deposited at the bottom of water, while mud is a mixture of water and combination of soil, silt, and clay. Earthquake occur strongly when Earth surface is geologically water-saturated like silt and mud. It can cause buildings to tilt,

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		underground pipes to break, infrastructures to be displaced and many more during earthquake.
56	D	Earth's solid core is composed of iron with lesser amounts of solid nickel. Layers 1 – 3 contains the following: ▪ 3 – liquid outer core is composed of iron mixed with nickel and trace amounts of lighter elements. ▪ 2 – lower mantle rich in iron and magnesium ▪ 1 – lower part of the crust is composed mainly of basalt and granite
57	B	Convection currents along the asthenosphere happens when warm air is less dense than colder air, and so warm air rises, as it rises it expands and cools, and then begins to fall. The heat is transferred to the surface where less dense magma separates the plates at the divergent zone (the Mid-Atlantic ridge. When the plates move away from the spreading centers, they cool, and the higher density basalt rocks that make up ocean crust formed at the ocean trenches/subduction zones. The crust is recycled back into the asthenosphere. A and C are other mechanisms of heat transfer not shown in the diagram D is called Incoming Solar Radiation.
58	B	The answer B corresponds to the permeability (from the greatest permeable to the least permeable) characteristic of the material
59	D	Convection currents happens along the asthenosphere or on the upper mantle happens when warm air is less dense than colder air, and so warm air rises, as it rises it expands and cools, and then begins to fall.
60	A	Oldest to youngest = bottom to top sequence.

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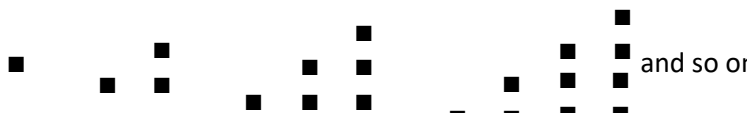
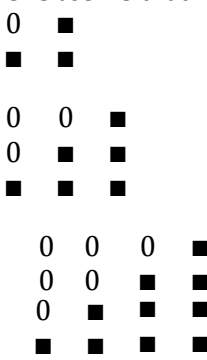
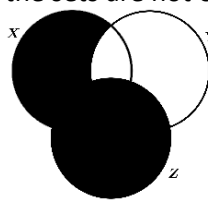
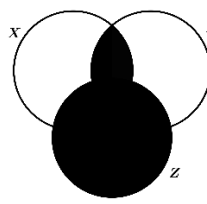
MATHEMATICS

ITEM NO.	ANSWER	RATIONALIZATION
1	A	Each element in the domain (real numbers) of the function defined by corresponds to a unique image (real numbers). This can further be verified by sketching the graph of $f(x)$ and check that it is a one-to-one function by using the horizontal line test. Horizontal Line Test: A function is one-to-one if and only if each horizontal line intersects the graph of the function in at most one point.
2	C	The diagram below satisfies the correct set of true statements. <pre> graph TD Q[QUADRILATERAL] --> P[PARALLELOGRAM] P --> R[RECTANGLE] P --> Rh[RHOMBUS] R --> S[SQUARE] Rh --> S </pre>
3	D	“If two sides of a quadrilateral are parallel and congruent, then the quadrilateral is a parallelogram.” P: Two sides of a quadrilateral are parallel. Q: Two sides of a quadrilateral are congruent. R: The quadrilateral is a parallelogram. <i>If two sides of a quadrilateral are</i> $\overbrace{(P \wedge Q)}^{(P \wedge Q)} \rightarrow$ <i>parallel and congruent, then the</i> <i>quadrilateral is a</i> $\overbrace{\text{parallelogram}}^R$.
4	B	$\sin \beta = -\frac{3}{5} \quad \cos \beta = -\frac{4}{5}$ $\cos^2 \beta - \sin^2 \beta = \left(-\frac{4}{5}\right)^2 - \left(-\frac{3}{5}\right)^2$ $= \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$
5	A	$B \cup C$ $(B \cup C)^c$ A $A \cap (B \cup C)^c$

ANSWER KEY

6	D	The sequence defined in the problem is 1, 3, 5, ... Let $S_n = 225$. The formula in determining the number of rows is $S_n = \frac{n}{2} [2a_1 + (n-1)d]$ $225 = \frac{n}{2} [2(1) + (n-1)(2)]$ $450 = n(2 + 2n - 2)$ $450 = 2n^2$ $225 = n^2$ $15 = n$
7	C	$72 : P = 3 : 2$ $3P = 144$ $P = 48$ $protesters : policemen = 12 : 1$ $protesters : 48 = 12 : 1$ $protesters = 48(12)$ $protesters = 576$
8	D	$nPr = 6P6 = 6! = 720$
9	C	There are $\binom{10}{2} = 45$ ways of drawing 2 balls from the box of 10 balls. To get an even sum, the two balls must have labels that are both odd. Among the ten prime numbers less than 30, 9 of them are odd (and the other is 2, which is even). This means that there are $\binom{9}{2} = 36$ ways of drawing 2 balls with even sum. Therefore, the required probability is $\frac{36}{45} = \frac{4}{5}$.
10	A	We check the ratio of two consecutive terms in each choice. A. $\frac{abrs}{ab} = rs, \frac{abr^2s^2}{abrs} = rs, \frac{abr^3s^3}{abr^2s^2} = rs, \dots \dots a \text{ geometric sequence}$ B. $\frac{abr}{ab} = r, \frac{abs}{abr} = \frac{s}{r} = 1, \frac{abr^2}{abs} = \frac{r^2}{s} = r, \frac{abs^2}{abr^2} = \frac{s^2}{r^2} = 1, \dots \text{NOT a geometric sequence always}$ C. $\frac{a+b+r+s}{a+b} \neq \frac{a+b+r^2+s^2}{a+b+r+s} \dots \dots \text{NOT a geometric sequence always}$ D. $\frac{a-b+r-s}{a-b} \neq \frac{a-b+r^2-s^2}{a-b+r-s} \dots \dots \text{NOT a geometric sequence always}$
11	D	From the information above, we obtain the following system of equations: $\begin{cases} a + b = 2 \\ 3a - 2b = 16 \end{cases}$ The solution of the system is $a = 4, b = -2$.
12	B	Given: $P = 80,000, r = 0.035, t = 1, I = \text{interest} = ?$ $I = Prt = (80,000)(0.035)(1) = 2,800$
13	C	Let x = the amount allotted for the internet café and y = the amount allotted for the photo booth. From the information above, we obtain the system $\begin{cases} x + y = 120,000 \\ 0.05x + 0.04y = 5,800 \end{cases}$ where solution is $x = 80,000$ and $y = 40,000$.
14	B	
15	D	
16	D	
17	C	
18	B	
19	D	
20	C	

ANSWER KEY

21	A	
22	D	
23	D	By the triangle inequality theorem, we have the following inequalities for the sides • $13+17>z$, so $30>z$ • $z+13>17$, so $z>4$ • $z+17>13$, so $z>-4$ From the inequalities, the only possible values for z are between 4 and 30. So 29 is a possible value of the z.
24	D	I is true because 55 is the 10th triangular number. II is true. The numbers can be represented as the following figures  Observe that from each figure, you obtain the triangular number of by adding the first n numbers. Hence $t_n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ III is also true. Observe that $1+3=4$ $3+6=9$ $6+10=16$  And so on.....
25	C	We can check each equality using Venn diagram. Among the three choices, the left-hand-side and right-hand-side sets of choice C are represented by different regions in the Venn diagram, as shown separately below, and so the sets are not equal.  $(X - Y) \cup Z$  $(X \cup Z) \cap (Y \cup Z)$
26	B	
27	C	
28	B	
29	C	
30	D	
31	A	
32	B	
33	A	
34	B	

ANSWER KEY

35	A	$f(x) = \sqrt{x^2 - 1}$ $x^2 - 1 \geq 0$ $(x - 1)(x + 1) \geq 0$ <table border="1"><tr><td></td><td>$x < -1$</td><td>$-1 < x < 1$</td><td>$x > 1$</td></tr><tr><td>$x - 1$</td><td>-</td><td>-</td><td>+</td></tr><tr><td>$x + 1$</td><td>-</td><td>+</td><td>+</td></tr><tr><td></td><td>+</td><td>-</td><td>+</td></tr></table> From the table above (consider it as an imaginary number line), for any values less than or equal to -1 and greater than or equal to 1 $f(x)$ is non-negative. Thus, the solution is denoted by $(-\infty, -1] \cup [1, +\infty)$.		$x < -1$	$-1 < x < 1$	$x > 1$	$x - 1$	-	-	+	$x + 1$	-	+	+		+	-	+
	$x < -1$	$-1 < x < 1$	$x > 1$															
$x - 1$	-	-	+															
$x + 1$	-	+	+															
	+	-	+															
36	A	"If the diagonals of a quadrilateral bisect each other and they are perpendicular, then the quadrilateral is a rhombus." P: The diagonals of a quadrilateral bisect each other. Q: The diagonals of a quadrilateral are perpendicular. R: The quadrilateral is a rhombus.																
37	A	The sequence defined in the problem is 3, 5, 7, ... Let $S_n = 120$. The formula in determining the number of rows is $S_n = \frac{n}{2} [2a_1 + (n - 1)d]$ $120 = \frac{n}{2} [2(3) + (n - 1)(2)]$ $240 = n(6 + 2n - 2)$ $240 = 2n^2 + 4n$ $2n^2 + 4n - 240 = 0$ $n^2 + 2n - 120 = 0$ $(n - 10)(n + 12) = 0$ $n = 10$																
38	A	$nPr = {}_{10}P_4 = 5040$																
39	C	From the information above, we obtain the following system of equations: $\begin{cases} a + b = -3 \\ -a + 2b = 18 \end{cases}$ The solution of the system is $a = -8, b = 5$.																
40	B	Let x = grams of nuts and y = grams of green peas. From the information above, we obtain the system $\begin{cases} x + y = 1,000 \\ 0.05x + 0.08y = 62 \end{cases}$ whose solution is $x = 600$ and $y = 400$.																

ANSWER KEY

MECHANICAL-TECHNICAL

ITEM NO.	ANSWER	RATIONALIZATION
1	B	The direction of force exerted by the truck on the crate would be the same as the direction of the truck's motion.
2	D	Sound travels fastest through solids because the particles are closest together. It travels less quickly in gases because the particles are further apart.
3	B	$5 \mu\text{A}(10 \text{ secs}) = 5 \times 10^{-5} \text{ coulombs}$
4	C	Changing gears on a ten-speed bicycle is a good example of using different size gears to change speed.
5	B	The picture shown is that of a socket wrench. From the choices given, a hex-head bolt which is tightened and loosened by a socket wrench is more relative.
6	A	Gear C is 4 times the diameter of gear A. Since the gears are all connected by a chain, the tip velocity of all the gears must be the same; otherwise, the chain would come off the gears. Therefore, if the tip velocity is to be the same for all gears and gear C is 4 times larger than gear A, then gear C must be turning 4 times slower than gear A. Twenty RPM divided by 4 equals 5 RPM.
7	A	Gravitational potential energy depends upon height ($PE=m \cdot g \cdot h$). A. Since point A is located at a point higher than point B, the potential energy at A is greater than the potential energy at B.
8	C	
9	C	Resonance happens when one object vibrating at the same natural frequency of a second object forces that second object into vibrational motion.
10	B	In a projectile motion, the reach is defined by the equation: $x = V_i (\cos \theta) t$; whereas the time of flight is defined by the equation: $t = 2V_i (\sin \theta) / g$. Where V_i is the initial velocity, the angle θ is the initial angle of projectile and g is the gravitational constant. Simplifying $x = V_i (\cos \theta) * 2V_i (\sin \theta) / g$ Since $2 (\cos \theta) (\sin \theta) = \sin 2\theta$ ----- double angle trigonometric identity Therefore: $x = V_i^2 (\sin 2\theta) / g$ For angle θ less than 45° the value of $(\sin 2\theta)$ is increasing as you increase the value of the angle. But for angle θ is greater than 45° but less than 60° the value of $(\sin 2\theta)$ is decreasing. For initial velocity V_i : the range x is increasing as it is the square of the initial velocity. The farthest distance is projectile X because the initial velocity is twice the value of the other even though the angle remains the same.
11	A	The bigger the driving wheel (rear wheel) the farther the distance it can travel with one rotation of the pedal because the circumference is larger compared to a smaller driving wheel, regardless of the size of the front wheel since it is only freewheeling.

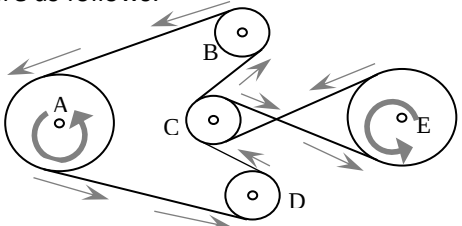
ANSWER KEY

		There are two bicycles with bigger driving wheel (bicycles W and Z), therefore the correct answer is A.
12	B	In Torricelli's Theorem, the velocity of a discharging fluid on a container is defined by the equation $v = \sqrt{2gh}$, where g is the gravitational constant and h is the height fluid or the vertical distance between the fluid surface and the point of discharge. Therefore, the lower the location of the holes, the higher the discharge velocity of fluid. Then the possible situation is B.
13	A	The material with higher thermal conductivity will have the highest temperature on the top surface given that all blocks are introduced to the same amount of heat from the burning wood. From the four materials, copper has the highest thermal conductivity, since it is being used as a heating element in most of the devices that produces heat such as flat iron, electric stove and water heater. Whereas brick has the lowest thermal conductivity among the four materials. Since it is being used to dissipate heat in furnaces or fireplaces. Aluminum has a higher thermal conductivity compared to steel. Therefore, the answer is A.
14	D	
15	B	
16	C	
17	B	Seeing lightning before hearing thunder shows us the light travels faster than sound.
18	A	This tool (chisel) carves wood. HAMMERS are used to drive nails; WELDERS or TORCHES are used to weld metal; WRENCHES are used to tighten bolts.
19	B	Choice IV is a picture of a carpenter's level which is used to determine whether a structure is built level to the ground while choices I, II and III are all common electrician tools. I. Wire Stripper II. Multimeter III. Flat-blade screwdriver
20	C	Gear C and D are turning clockwise. The other gears are turning counterclockwise. It helps to follow the direction of the chain, which is connected to all of the gears.
21	B	Gravitational potential energy depends upon height ($PE = m \cdot g \cdot h$). The PE is a minimum when the height is a minimum. Position B is the lowest position in the figure.
22	C	Using the formula for density $\rho = \frac{m}{V}$, with appropriate units of representation.
23	C	As defined, a plane is a metal tool with a handle and an adjustable blade, used to shave off thin strips of wood for smoothing or leveling.
24	B	The tendency to increase or decrease the wavelength of the sound due to the motion of the source is called the Doppler effect. Hence, when the Fire truck is coming toward you, the wavelength is decreased, and the pitch is above the pitch when the source is at rest. When the truck passes and is moving away from you, the opposite holds true. The

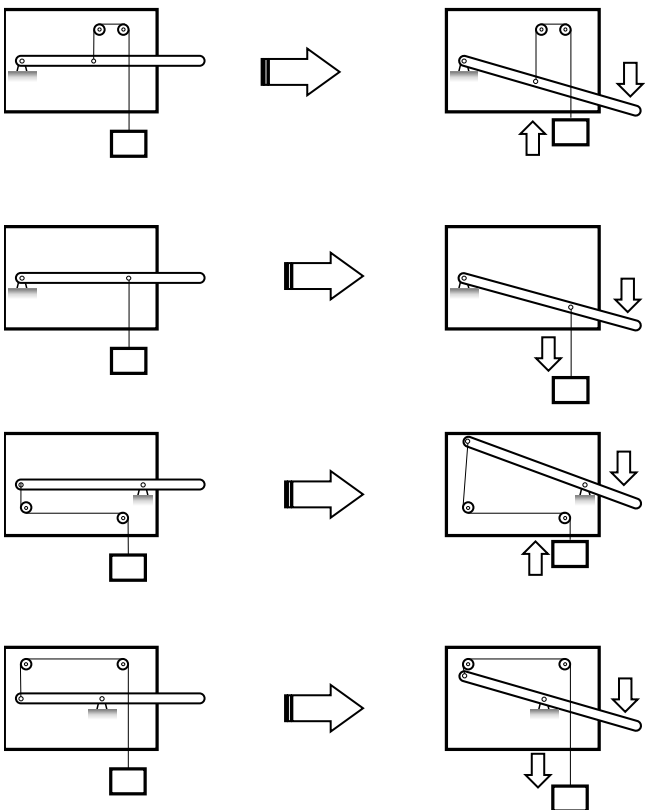
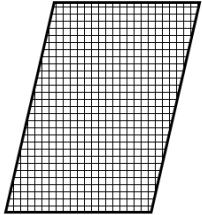
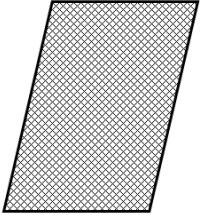
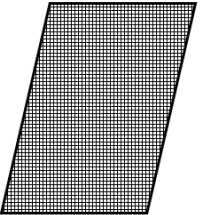
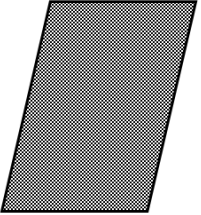
ANSWER KEY

		wavelength is increased, and the pitch is decreased below the value when the source is at rest. The result is the familiar change of pitch from high to low as the Fire truck passes.
25	B	In a projectile motion, the reach is defined by the equation: $h_{\max} = [V_i (\sin \theta)]^2 / 2g$; Where V_i is the initial velocity, the angle θ is the initial angle of projectile and g is the gravitational constant. Simplifying $h_{\max} = V_i^2 (\sin \theta)^2 / 2g$ For angle θ less than 90° the value of $(\sin \theta)^2$ is increasing as you increase the value of the angle. For initial velocity V_i: the maximum height is increasing as it is the square of the initial velocity. The highest point is reached by projectile X because the initial velocity is twice the value compared to other projectiles even though the angle remains the same.
26	A	P_H = pressure of the atmosphere or P_{ATM}, since it is exposed to the atmosphere P_F = pressure of the atmosphere or P_{ATM}, since it is also exposed to the atmosphere $P_D = P_F + \text{Pressure due to the weight of the oil} = P_{ATM} + \gamma_{oil} h_{oil(D-F)}$; where γ_{oil} is the specific weight of oil and $h_{oil(D-F)}$ is the height of oil from points D to F $P_B = P_F + \text{Pressure due to the weight of the oil} = P_{ATM} + \gamma_{oil} h_{oil(B-F)}$; where γ_{oil} is the specific weight of oil and $h_{oil(B-F)}$ is the height of oil from points B to F $P_{BOTTOM} = P_B + \text{Pressure due to the weight of the water} = P_B + \gamma_{water} h_{water(Bottom-B)}$ $= P_{ATM} + \gamma_{oil} h_{oil(B-F)} + \gamma_{water} h_{water(Bottom-B)}$; where γ_{water} is the specific weight of water and $h_{water(Bottom-B)}$ is the height of water from bottom to point B $P_A = P_{BOTTOM} - \text{Pressure due to the weight of the water} = P_{BOTTOM} - \gamma_{water} h_{water(Bottom-A)}$ $P_A = P_{ATM} + \gamma_{oil} h_{oil(B-F)} + \gamma_{water} h_{water(Bottom-B)} - \gamma_{water} h_{water(Bottom-A)}$; where γ_{water} is the specific weight of water and $h_{water(Bottom-A)}$ is the height of water from bottom to point A since $h_{water(Bottom-A)} = h_{water(Bottom-B)}$ therefore: $P_A = P_{ATM} + \gamma_{oil} h_{oil(B-F)}$ $P_C = P_A - \text{Pressure due to the weight of the water} = P_A - \gamma_{water} h_{water(A-C)} = P_{ATM} + \gamma_{oil} h_{oil(B-F)} - \gamma_{water} h_{water(A-C)}$; where γ_{water} is the specific weight of water and $h_{water(A-C)}$ is the height of water from points A to C $P_E = P_C - \text{Pressure due to the weight of the air} = P_C - \gamma_{air} h_{air(C-E)} = P_{ATM} + \gamma_{oil} h_{oil(B-F)} - \gamma_{water} h_{water(A-C)} - \gamma_{air} h_{air(C-E)}$; where γ_{air} is the specific weight of air and $h_{air(C-E)}$ is the height of air from points C to E $P_G = P_C - \text{Pressure due to the weight of the air} = P_C - \gamma_{air} h_{air(C-G)} = P_{ATM} + \gamma_{oil} h_{oil(B-F)} - \gamma_{water} h_{water(A-C)} - \gamma_{air} h_{air(C-G)}$;

ANSWER KEY

		where γ_{air} is the specific weight of air and $h_{\text{air}(C-G)}$ is the height of air from points C to G Therefore, the only true statement is $P_A = P_B$ and the answer is A
27	A	In conduction of heat, the point closer to the thermal source has the higher temperature. Since the source of heat is air at 100°C on both sides exposing the outer surface of the two materials, therefore points A and D are the possible points that have higher temperature compared to points B and C. Point A is on the steel side that has a higher thermal conductivity compared to brick. That will give us point A with the highest temperature compared to the other points.
28	D	The rotation of pulley at A is counterclockwise. The movements of the belts are as follows.  Then the rotation of the pulley at E is also counterclockwise. Regarding the speed of pulley at E relative to pulley at A: First, we relate the speed of pulley at C relative to pulley at A. Since pulley at C is half the size of the pulley at A, then the speed of the pulley at C is twice the speed relative to pulley at A. $\text{Speed}_C = 2 \text{ Speed}_A$ Relating pulley at E relative to pulley at C, since the pulley at E is twice the size of the pulley at C, then the speed of the pulley at E is half the speed relative to pulley at C. $\text{Speed}_E = \frac{1}{2} \text{ Speed}_C$ Hence, $\text{Speed}_E = \frac{1}{2} (2 \text{ Speed}_A)$ $\text{Speed}_E = \text{Speed}_A$ Therefore, the answer is D.

ANSWER KEY

29	B	 Therefore, the answer is B.
30	A	Given different sizes of sand grains (from gravels or rocks to the finest sand); A is big rocks, B is small rocks, C is course sand, D is fine sand, and E is finest sand. For the following wire mesh to sieve granules of sand: below are the list of sizes than can pass through the mesh.  Mesh W B, C, D and E  Mesh X C, D and E  Mesh Y D and E  Mesh Z E only A. W – X – Y – Z Mesh W will separate A and the rest will pass through to the next stage of sieving. On the next stage, mesh X will separate B and the rest will pass through to the next stage. Next is mesh Y will separate C the rest will pass through to the last stage of sieving. Lastly, mesh Z will separate D and E will be the final output of the filtering. B. Z – Y – X – W

ANSWER KEY

	Mesh Z will only filter E and the rest will stay at the top of the mesh. In which the that sand compile on the top of mesh Z cannot proceed to the next stage of sieving that results to only finest sand can be separated. C. W – Z – Y – X Mesh W will separate A and the rest will pass through to the next stage of sieving. On the next stage, Mesh Z will only filter E and the rest will stay at the top of the mesh. In which the that sand compile on the top of mesh Z cannot proceed to the next stage of sieving that results to rocks and finest sand are separated from the other granules. D. Z – X – Y – W Mesh Z will only filter E and the rest will stay at the top of the mesh. In which the that sand compile on the top of mesh Z cannot proceed to the next stage of sieving that results to only finest sand can be separated. Therefore, the answer is A.
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